# Louis Hildebrand

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## Education

**McGill University** MSc Electrical Engineering *Winter 2024–Winter 2026*
- **GPA:** 4.0/4.0
- **Supervisor:** Prof. Christophe Dubach (Compilation and Architecture Synthesis Lab)
- **Thesis:** "A Minimal Intermediate Language for Generating Streaming Accelerators"

**McGill University** B. Software Engineering *Fall 2020–Fall 2023*
- **GPA:** 4.0/4.0
- Dean's Honour List: 2020/2021, 2021/2022, 2022/2023
- British Association Medal (highest final exam grades)

**John Abbott College** Honours Science *Fall 2018–Winter 2020*
- Valedictorian
- Dean's List: Fall 2018, Winter 2019, Fall 2019

## Publication

L. Hildebrand and C. Dubach, "Sirop: A small IR for HLS with parallel patterns," in *Proceedings of the 27th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES '26)*

Paper Talk Code Docs

## Skills

- **Formal languages:** Scala, Python, VHDL, C, Assembly (ARMv7, MIPS), Rust, Java, C#, SQL (MS SQL Server, PostgreSQL), OCaml, JavaScript, HTML, CSS, etc.
- **Natural languages:** English, French, Afrikaans
- **Frameworks:** Spring Boot, .NET (Framework, Core), Django, Vue.js
- **Other tools:** Git, Bash, Make, Valgrind, GitHub Actions CI, Gradle, JUnit, etc.
- **Operating systems used:** Ubuntu, Windows

## Work Experience

**McGill; ATLAS Experiment (CERN)** Research Assistant *April 2026–April 2027*
- FPGA acceleration for the Phase-II upgrade of the Liquid Argon Calorimeter
- Developing a compiler to generate resource-efficient FPGA implementations of neural networks for energy reconstruction

**MDA Space** Engineering Intern, DevOps *Summer 2023*

- Optimized SQL queries and stored procedures, achieving up to $\sim 10\times$ speedups
- Implemented new features and fixed bugs in web services using ASP.NET MVC, Razor Pages, Telerik, and Kendo UI

**123Loadboard** Backend Intern *Summers 2021, 2022*

- Independently implemented new microservices with C# and .NET Core given a predefined specification
- Fixed bugs in and added new endpoints to the main API (.NET Framework)

## McGill Teaching Assistant Experience

**Computer Organization** (ECSE 324) *Fall 2025*

- Delivered tutorials on computer organization (e.g., interacting with devices via memory-mapped I/O)
- Guided students in lab work (writing C and ARM assembly programs)
- Answered students' questions on the online discussion board
- Graded assignments

**Model-Based Programming** (ECSE 223) *Winter 2025*

- Delivered weekly tutorials on model-based programming (e.g., UML class and state diagrams, Umple) and other tools (e.g., Git, JUnit, Gradle, Cucumber)
- Answered students' questions on the online discussion board
- Helped prepare assignments

**Intro. to Software Engineering** (ECSE 321) *Fall 2022–Winter 2025*

- Delivered weekly tutorials on developing a fullstack web app with PostgreSQL, Spring Boot, and Vue.js
- Held weekly office hours and answered questions on the online discussion board
- Helped write and grade tests

**Ordinary Differential Equations for Engineers** (MATH 263) *Fall 2021*

- Delivered weekly tutorials on differential equations, including a summary of lecture content and practice problems
- Answered students' questions by email
- Graded assignments

## Selected Courses

**Language-Based Security** (COMP 523) *Winter 2025*

- Formally studied the syntax and semantics of programming languages and type systems
- **Project:** `chick`, a type checker for a dependently-typed language

**Computer Graphics** (ECSE 532) *Fall 2024*

- Learned the fundamentals of computer graphics: 3D transformations, meshes, the graphics pipeline, lighting, textures, etc.
- **Project:** a raytracer with support for surfaces of different colour and smoothness, mirrors, depth of field blur, spherical environment maps, textures, etc. Implemented in Python using the taichi library for GPU acceleration.

**Machine Learning for Engineers** (ECSE 551) *Fall 2024*

- Studied classical machine learning models (decision trees, naïve Bayes, etc.) as well as neural networks, CNNs, and RNNs
- **Project:** stacked classifier (with random forest, logistic regression, etc. as the base models) to categorize Reddit posts from four cities. Achieved the second-highest accuracy on the test dataset (out of 25 groups).

**Compiler Design** (COMP 520) *Winter 2024*

- Learned to implement a compiler, including parsing, semantic analysis, register allocation, and code generation
- **Project:** compiler that translates a subset of C to MIPS assembly

**Microprocessors** (ECSE 444) *Fall 2023*

- Programmed an STM32 B-L4S5I-IOT01A board using C and ARMv7 assembly
- **Project:** memory game that plays a series of tones (high or low), detects user inputs via accelerometer (up or down), and provides feedback via a speaker

**Parallel Computing** (ECSE 420) *Fall 2023*

- Learned GPU programming with CUDA
- **Project:** CUDA implementation of a general 2D cellular automaton simulator, achieving 590× higher throughput than an equivalent sequential implementation in C

**Operating Systems** (ECSE 427) *Fall 2022*

- Learned fundamental OS concepts: processes, threads, memory management, etc.
- **Assignments:** a simple shell, threading library, and file system (all in C)

# Projects

**Twisty Timer** (Java Android app)

- Contributed new features to an existing Rubik's Cube app (e.g., a practice mode for blindfolded solving)

**SH Prediction** (Python command-line app)

- Predicts players' roles in the social deduction game "Secret Hitler"

**Pocket Cube Solver** (Arduino project)

- Robot to solve a 2×2×2 Rubik's Cube
- Presented at the 2018 Montreal Regional Science and Technology Fair
- **Prizes:** Intel Excellence in Computer Science Award, McGill University School of Computer Science (Robotics) Award